

Team Listen

Does the Team Actually Listen? A Causal Audit of Language Grounding in Language-Conditioned Multi-Robot Policies

Research and publication plan · September 2026 · repo: github.com/ShashikaHDS/team-listen · fully simulation-based (Isaac Lab, single RTX 5090) · target: one journal paper in 6–12 months

1 Executive summary

Robots that follow natural-language commands are the current center of gravity in robot learning. In 2025–2026, careful audits of single-arm systems (LIBERO-Plus, LangGap) revealed an uncomfortable fact: the policies largely *ignore* their instructions and solve tasks from visual context alone, with nobody noticing because the guess is usually right. No such audit exists for robot *teams*, even though teams are where language does irreplaceable work: assigning roles between identical-looking targets, imposing precedence, expressing exclusion. Every current multi-robot language system reports only aggregate task success, which is precisely the condition under which a dead language channel stays invisible.

This project builds the measurement instruments that make language grounding in multi-robot policies a measured quantity rather than an assumption: a task family where language is *provably* necessary, a metric (the Coordination Sensitivity Index) that survives multi-agent stochasticity, and a controlled comparison of four ways of feeding language to a team. The deliverable is one journal paper, targeted at IEEE Transactions on Artificial Intelligence, with the complete harness released open-source.

2 What we are doing

We train cooperative multi-robot policies with multi-agent reinforcement learning (MAPPO) in NVIDIA Isaac Lab, condition them on natural-language instructions through four different interfaces, and then ask a causal question: **if we change only the instruction, holding the world and all randomness fixed, does the team's behaviour change?** Around that question we build:

- 1. A language-necessary task family.** Cooperative tasks (role binding over identical stations, precedence ordering; later exclusion and mid-episode reassignment) engineered so geometry alone cannot reveal the correct assignment. Each task ships a machine-checked *language-necessity certificate*: a full-state but instruction-blind "oracle" policy trained to convergence must sit exactly at chance on assignment. If it beats chance, the task leaks and is redesigned. The certificate is backed by a paired evaluation design in which a blind policy's accuracy is 0.5 *by algebraic identity*, checked bit-for-bit by machine, plus a supervised leak probe over the observations.
- 2. The Coordination Sensitivity Index (CSI).** Behavioural divergence under a counterfactual-instruction intervention, normalised by the divergence a physics-seed change alone produces. The normalisation is what makes counterfactual auditing portable to multi-agent systems, whose intrinsic stochasticity would otherwise swamp raw behavioural deltas. Reported per semantic category with episode-clustered bootstrap confidence intervals.
- 3. A four-way conditioning comparison.** Global instruction broadcast, per-agent atomic prompts, a frozen vision-language-model commander emitting per-agent sub-goal tokens, and a frozen sentence embedding — all trained under identical MARL and audited with the same instruments, including whether CSI predicts generalisation to unseen instruction combinations and team sizes.

Pre-registered headline bet: teams will prove language-sensitive on *role binding* but nearly inert on *precedence*, because emergent collision avoidance satisfies ordering constraints for free. A team can be provably deaf to half its instruction while scoring high aggregate success — a dissociation invisible to every existing evaluation.

3 Why: the gap and why now

The direction was selected by a systematic gap study: six literature sweeps (VLA models, MARL for robotics, VLM-based coordination, the Isaac ecosystem, venue trends, evaluation critiques) generated nine candidate directions, each then subjected to an adversarial novelty check that actively searched for existing work filling it. Three candidates were eliminated by prior work; this direction survived with the largest verified delta and the highest feasibility score for our resources.

- **The single-arm audit line stops at one robot.** LIBERO-Plus and LangGap perturb object-naming language only; role binding, precedence, exclusion, and reassignment are structurally inexpressible with one arm.
- **The multi-robot language line never audits.** CHORUS, MA-VLA, CoVLA and the competition entrants report aggregate success only, and all coordinate through a shared workspace view — the setting where an inert language channel is undetectable.
- **No metric exists** for instruction sensitivity that survives MARL stochasticity, and no construct-validity device exists for "this task needs language" claims anywhere in the literature.
- **Why us:** the method needs no giant-model training (everything language-side is frozen; only small MARL policies train, well within one RTX 5090), it reuses our proven assets — the verified grid fleet environment from the rendezvous paper, the Isaac Sim evaluation pipeline, and a seeded paired-statistics evaluation discipline — and the audit-paper shape rewards exactly that statistical rigor.

4 How: method and infrastructure

Stack. Isaac Lab DirectMARLEnv (PettingZoo parallel API) over a zero-asset, pure-tensor grid world ported from our rendezvous environment with machine-verified transition parity; skrl MAPPO with discrete actions; 8192 parallel environments on one RTX 5090. Language embeddings are computed once offline (MiniLM) and cached; the VLM commander is distilled to a symbolic sub-goal vocabulary for training, with the real 3-billion-parameter VLM entering only at evaluation. A cross-fidelity robustness arm re-evaluates final policies under RTX rendering, raycast sensing, and localisation noise.

Statistical protocol (pre-registered in the repository before any evaluation): equivalence testing (TOST) on hierarchical bootstrap intervals that resample scenarios and seeds jointly; episodes, never agents, as the clustering unit; paired counterfactuals sharing identical physics seeds; Holm correction across perturbation families; a stated power profile.

Milestone	Content	Gate
M1 (months 1–1.5)	Two task variants, certificate oracle, first language-conditioned policy, first CSI measurement	Certificate at chance AND near-zero CSI on precedence → GO; high CSI → pivot to architecture-comparison framing
M2 (2–3)	All four conditioning interfaces, CSI v1 table, dissociation probe	The paper's identity is fixed here
M3 (4–5)	Full task family (exclusion, reassignment), N=2–4, baselines	Perturbation protocol frozen
M4 (6–7)	CSI-predicts-generalisation study; repair-control arm	Second headline plot; scope freeze, arXiv v1
M5 (8–9)	Cross-fidelity robustness arm	Bounded; droppable without harming the core claim
M6 (10–12)	Harness release, compression, submission	Journal submission

5 The paper

Venue. A five-way evidence-backed venue study ranked **IEEE Transactions on Artificial Intelligence** first (fit 8/10): it is the only IEEE transactions with published precedent for evaluation-methodology papers without a new algorithm, imposes no hardware expectation (we are fully simulated), and returns first decisions in roughly twelve weeks. The framing requirement: submit as *measurement methodology* — the metric leads, the robots are the substrate. A pre-submission enquiry to the new IEEE Transactions on Robot Learning at month 5 is the one cheap hedge (best topical fit, but its guidelines expect hardware; a written yes flips it to primary). Backup chain: TCDS → TNNLS (with added representation analysis) → TMLR. We deliberately avoid T-RO and RA-L (hardware-trained reviewer pools, page limits) and post an arXiv preprint at month 7 to hold priority in a fast-moving subgenre.

Contributions as the paper states them: (1) the certified task family and the certificate as a construct-validity device with no precedent in either audit line; (2) CSI with formal properties (boundedness, invariances, degenerate-case handling) and a finite-sample estimator analysis; (3) the pre-registered role-binding/precedence dissociation; (4) the four-way interface comparison with predictive-validity evidence and the open harness. Writing proceeds alongside the experiments: the pre-registration document doubles as the first draft of the formalisation sections, and the metric specification is written before the headline results exist.

6 Status (1 September 2026)

Area	State
Direction & novelty	Adversarially verified against 2024–2026 literature; decision log maintained in-repo
Implementation spec	Red-teamed by three independent reviews; 24 critical defects fixed pre-code (including a certificate design that was provably vacuous and a guaranteed deadlock in the precedence task)
Code & tests	Environment core with bit-parity to the rendezvous reference; rewards with machine-checked instruction-purity; scenario banks; rollout harness with cross-process bit-identical paired lanes; certificate, leak probe, statistics toolkit, CSI — 17 test files, all green

GPU validation	Throughput gate passed at $3.9\times$ the requirement (390k env-steps/s at 8192 envs; full M1 program ≈ 15 GPU-hours); audit machinery validated end-to-end on GPU (bitwise paired-lane equality at 4000 envs)
Training	First campaign exposed a mid-training instability of skrl MAPPO with discrete actions (adaptive-LR-driven logit explosion) — root-caused deterministically and fixed with a ratified LR ceiling. Completed policies were incompetent at the task, so the pre-registered λ -shaping pilot (the spec's planned response) is running now; the full campaign follows the first competent pilot cell
Pre-registration	Evaluation protocol constants frozen in git before any policy evaluation; OSF/arXiv registration is the month-1 writing task

7 Principal risks and mitigations

Risk	Mitigation
"Teams ignore language too" is pre-announced by the single-arm audits	The contribution is the certificate + CSI + dissociation + interface comparison, pre-registered; month 2–3 finds the differentiated finding or declares the alternative before further investment
Blind policies fail to learn the task at all	The pre-registered pilot ladder (shaping sweep $\rightarrow$ grid-size/density reduction) runs before the headline; a redesigned-task iteration is budgeted and itself reportable
Scooping in a monthly-preprint subgenre	Scope freeze at month 6, arXiv v1 at month 7; the specific delta (certificate + seed-normalised metric in a multi-agent setting) verified unoccupied
Venue rejects sim-only work	Primary venue chosen for exactly this tolerance; cross-fidelity arm answers the embodiment question; backup chain pre-agreed

Companion documents in the repository: RESEARCH_PLAN.md (full plan and prior-work analysis), PAPER_PLAN.md (venue evidence, section outline, writing calendar), M1_SPEC.md (implementation specification), DECISIONS.md (running decision log), plus the day-o, gate, training, and diagnosis reports.